# Midterm Practice B: Theory, Code Reading, and Checking AI

**Statistical Thinking for the AI Era - Spring 2027**

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| **Suggested time** | 75 minutes, followed by correction with the solution key |
| **Materials** | One student-prepared sheet, front and back; no calculator needed |
| **AI policy** | Complete once without AI. Afterward, use AI only to generate variants that you verify. |
| **Purpose** | Harder, scaffolded practice for theory, printed-code interpretation, and AI-assisted reasoning |

This is a second practice form, not a list of facts to memorize. Any hierarchical-model question used on a graded exam will supply the needed Beta-distribution facts and use the same scaffold. You will not be asked to write executable code, but you may be asked to read printed code.

## Part 1 - Statistical theory for familiar and flexible models (40 points)

### 1. A Beta-Bernoulli camera model (22 points)

Camera locations differ in how often animals appear. For camera $j$, consider

$$p_j \sim \text{Beta}(\alpha, \beta), \qquad Y_{ij} \mid p_j \sim \text{Bernoulli}(p_j),$$

where observations are conditionally independent within a camera given $p_j$. For this problem, $alpha = 2$ and $\beta = 3$.

You may use:

$$f(p) \propto p^{\alpha-1}(1-p)^{\beta-1}, \qquad \mathbb{E}[p] = \frac{\alpha}{\alpha+\beta}, \qquad \mathrm{Var}(p) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

a. Describe the two levels of randomness in this model. What becomes independent after conditioning on $p_j$? *(3)*

b. At one camera, $n_j = 5$ frames contain $s_j = 4$ animal-present outcomes. Show that the likelihood contribution for $p_j$ is proportional to $p_j^4(1-p_j)^1$. Multiply it by the prior kernel and identify the posterior distribution of $p_j$. *(5)*

c. Compute the prior mean, the observed proportion, and the posterior mean. Explain the posterior mean as partial pooling rather than as a replacement for the observed data. *(4)*

d. Consider two different frames from the same randomly selected camera. Use

$$\mathrm{Cov}(Y_{1j}, Y_{2j}) = \mathbb{E}\{\mathrm{Cov}(Y_{1j}, Y_{2j} \mid p_j)\} + \mathrm{Cov}\{\mathbb{E}(Y_{1j} \mid p_j), \mathbb{E}(Y_{2j} \mid p_j)\}$$

to show that their marginal covariance is $\mathrm{Var}(p_j)$. Compute that covariance. Then use $\mathrm{Var}(Y_{ij}) = 0.4(0.6) = 0.24$ to compute their correlation. Explain why treating all frames as independent would tend to understate uncertainty. *(7)*

e. Distinguish prediction for another frame at this observed camera from prediction for a frame at a new camera. Which sources of uncertainty differ? *(3)*

## 2. A hidden layer changes the decision region (18 points)

Let $g(z) = \max(0, z)$ be the ReLU activation. A one-hidden-layer classifier in one input dimension is

$$h_1(x) = g(x), \qquad h_2(x) = g(x-2),$$

$$s(x) = -1 + 2h_1(x) - 3h_2(x), \qquad \hat{y} = \mathbf{1}\{s(x) > 0\}.$$

a. Write $s(x)$ separately for $x \le 0$, $0 < x \le 2$, and $x > 2$. *(6)*

b. Find every decision boundary and the complete region classified as 1. State what happens exactly on a boundary under the stated strict inequality. *(5)*

c. Replace ReLU with the identity activation $g(z) = z$. Simplify $s(x)$, find the decision boundary, and describe the region classified as 1. *(5)*

d. In two sentences, explain how changing the activation changed the learned basis functions and the geometry of the decision region. *(2)*

## Part 2 - Read and check printed code (25 points)

The target is performance on later-period complaints, including repeated templates. The analyst prints the following code. You do not need to run it.

```
vectorizer = TfidfVectorizer(min_df=5)
X = vectorizer.fit_transform(dat["narrative"])
y = dat["issue_label"]

train_i, test_i = train_test_split(
    np.arange(len(dat)), test_size=.25, random_state=2027, stratify=y
)
model = LogisticRegression(max_iter=2000).fit(X[train_i], y.iloc[train_i])
test_prob = model.predict_proba(X[test_i])[:, 1]

candidate = [0.3, 0.5, 0.7]
test_accuracy = [accuracy_score(y.iloc[test_i], test_prob >= t) for t in candidate]
threshold = candidate[np.argmax(test_accuracy)]
reported_accuracy = max(test_accuracy)
```

a. Explain what lines 1-2, 5-7, and 10-12 are doing. Clearly distinguish fitting a representation, fitting a model, and selecting a decision threshold. *(6)*

b. Identify three threats to a defensible later-period performance claim. For each, point to the relevant line or omitted variable and state the likely consequence. At least one threat must concern information leakage and at least one must concern the sampling or dependence structure. *(9)*

c. Write a corrected analysis order in plain English or pseudocode. It must include a training set, a validation role, a final test role, representation fitting, template or time grouping, and exactly one final test evaluation. No Python syntax is required. *(6)*

d. If the analyst changes only `threshold` after the logistic regression has been fitted, which object changes and which object does not? If the threshold is selected using the test set, why can its reported performance still be optimistic? *(4)*

## Part 3 - Prompt, iterate, and evaluate the AI response (35 points)

A client wants to compare a documented logistic baseline with a vision system for detecting animals at new camera locations. The available file has 240 frames from 60 trigger sequences at 8 sampled cameras, a single human label per frame, and probabilities from both systems. Missing an animal is considered more consequential than sending an empty frame to review, but the relative cost has not been quantified.

### 3A. Write the initial prompt (10 points)

In at most 120 words, write a prompt asking an AI assistant to propose an evaluation plan. Your prompt must specify or request:

- the research question and new-camera target population;
- frame, sequence, and camera roles;
- the statistical baseline and AI system;
- a split or evaluation design;
- at least one performance quantity and the client decision;
- assumptions or missing information the assistant must flag;
- a prohibition against inventing results.

### 3B. Plan the iteration (8 points)

Name four issues that may require a follow-up prompt or a human decision after the first response. For each, explain why polished wording alone cannot settle it. At least one issue must involve measurement and at least one must involve the deployment decision.

### 3C. Check the returned answer (17 points)

Suppose the paired evaluation counts are:

| | Vision correct | Vision wrong |
|---|---|---|
| Baseline correct | 180 | 12 |
| Baseline wrong | 22 | 26 |

The assistant responds:

> "The vision system improves accuracy from 80.0% to 84.2%, a gain of 4.2 points. Treating the two accuracies as independent gives a standard error of about 3.5 points and a 95% interval from about -2.8 to 11.2 points, so the systems are equivalent. A random frame split and an individual-frame bootstrap are appropriate because frames are the model inputs. The 84.2% accuracy proves the system will work at new cameras, and threshold 0.5 should be deployed."

You may use $\sqrt{34} \approx 5.83$.

a. Check the two accuracies and their difference. Is the assistant's independent-proportions arithmetic approximately internally consistent, even though the analysis choice is wrong? *(4)*

b. Use the discordant counts to compute the paired standard error $\sqrt{b+c}/n$ and an approximate paired 95% interval. *(4)*

c. Identify four distinct errors in the assistant's statistical explanation. Numerical correctness and explanatory correctness must be evaluated separately. *(5)*

d. Rewrite the recommendation in at most four sentences. Include the baseline, paired estimate with uncertainty, evaluated population, unit/split limitation, decision limitation, and what must be learned before deployment. *(4)*